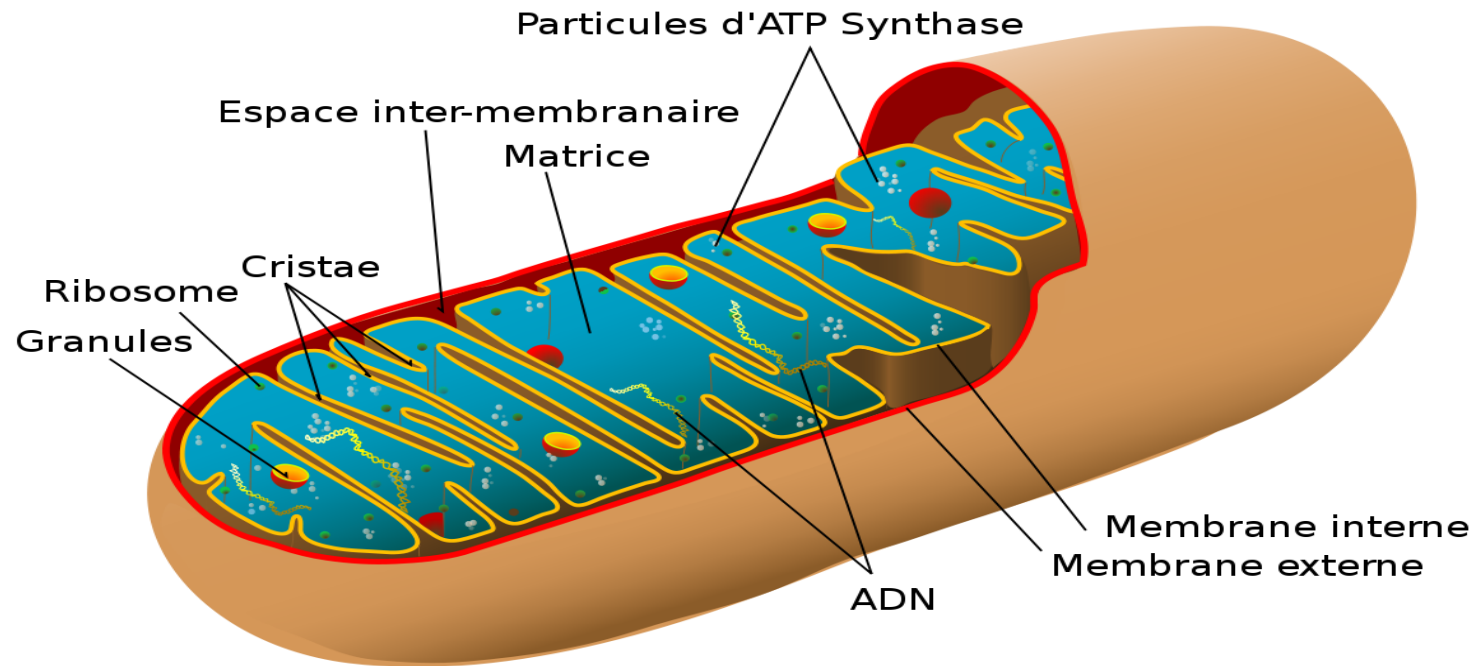


La mitochondrie





The outer membrane is a lipid bilayer 5 to 7 nm thick, with a composition similar to the plasma membrane but containing more protein, and is very rich in porins; it is permeable to ions and molecules with a molecular mass up to 6 kDa.

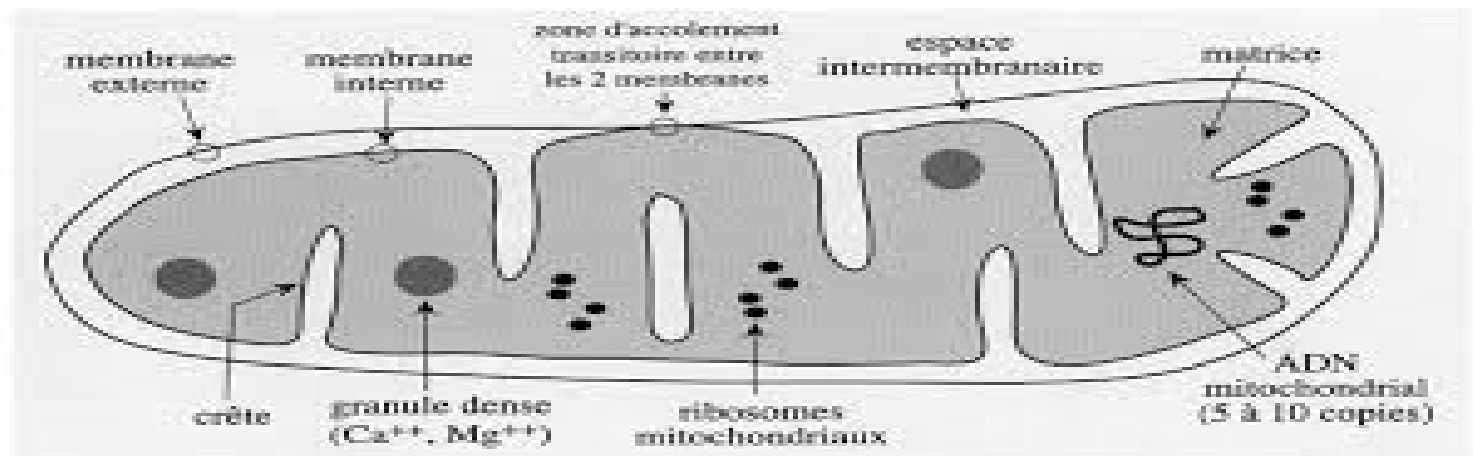
The intermembrane space is a dense space 4 to 7 nm thick, containing H^+ protons that play an important role in phosphorylation, cytochrome C molecules, and other molecules...



The inner membrane is a lipid bilayer, 5 to 6 nm thick, characterized by a very different organization from that of the outer membrane, as it is composed of 80% proteins and 20% lipids. It is very rich in transporters and enzymatic protein complexes, and it also contains TIM translocases involved in protein import.

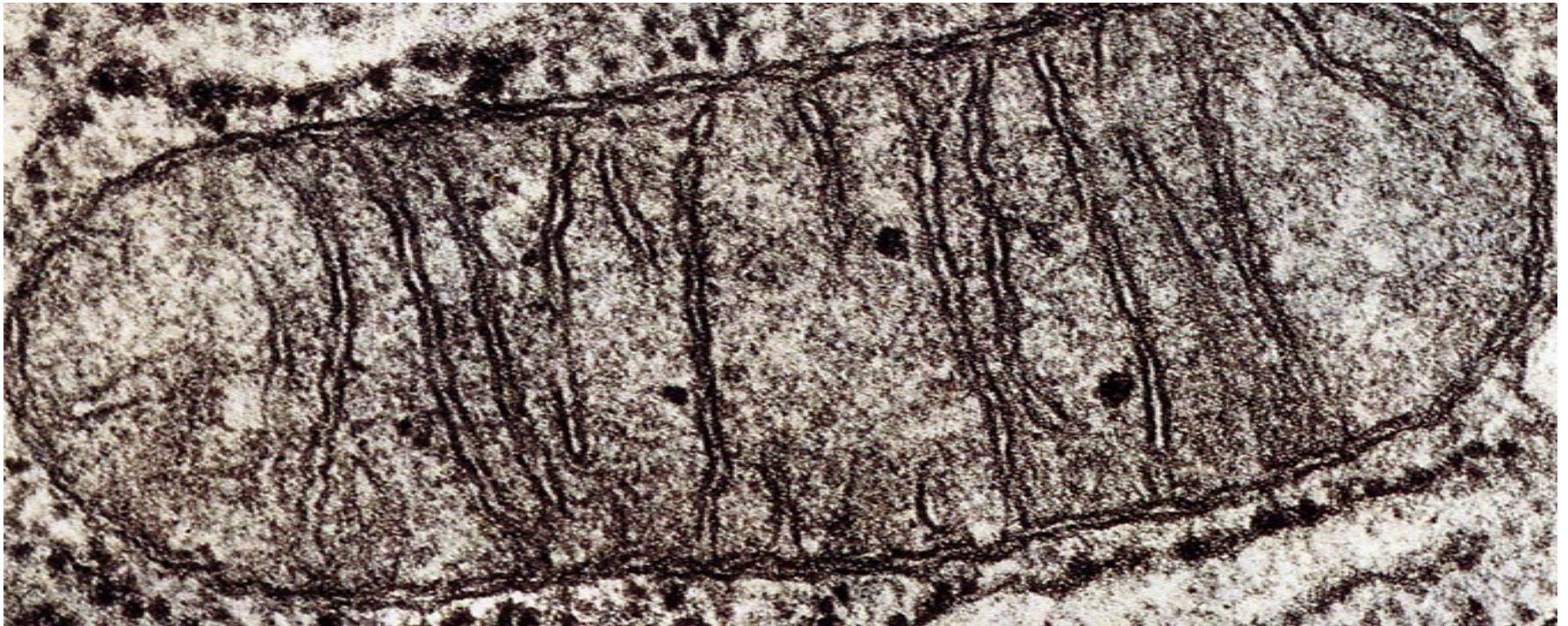
It has numerous folds (**Mitochondrial cristae**). This structure results in a threefold increase in its surface area compared to the outer membrane. The number and surface area of the cristae are correlated with the cell's ATP demand.

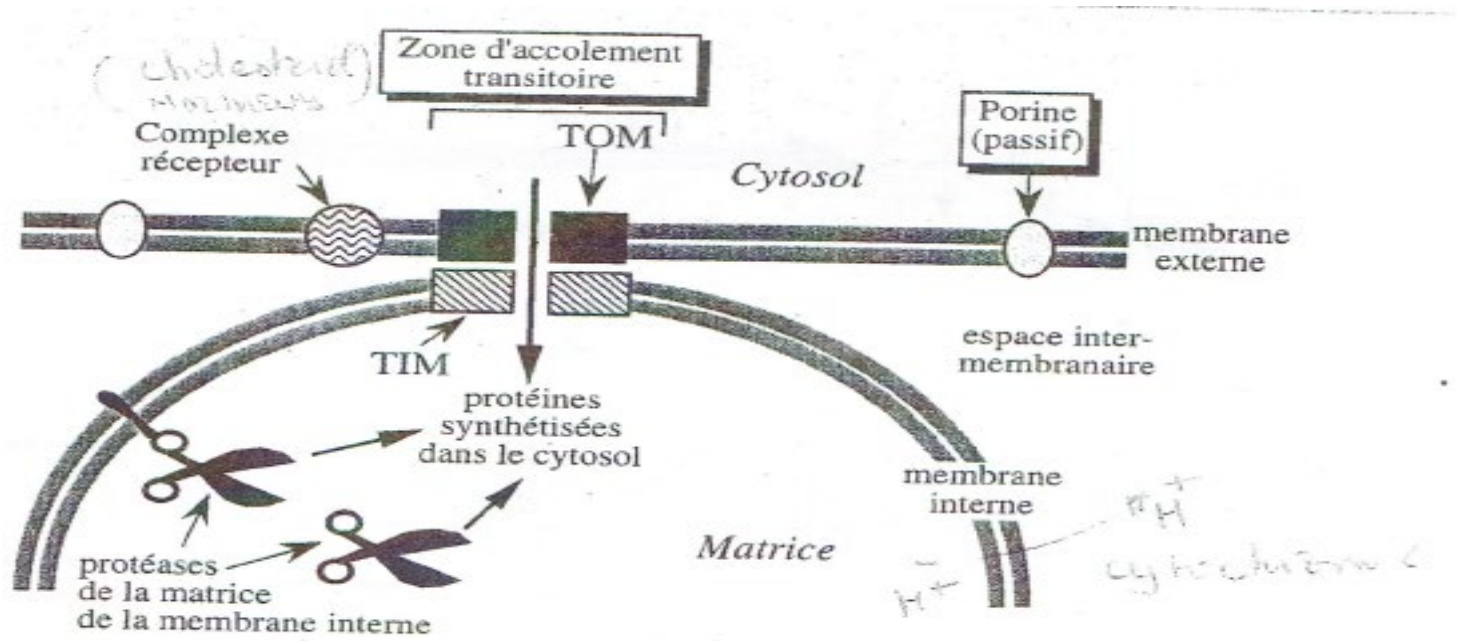
Transient binding sites These are sites of protein import from the cytosol into the mitochondria.



The mitochondrial matrix occupies the internal space and contains mitochondrial ribosomes, circular DNA molecules (mtDNA), messenger RNA and transfer RNA, dense and irregular granules formed by the accumulation of calcium and magnesium cations, and numerous enzyme systems.

Mitochondrial DNA Mitochondria possess their own genome (mtDNA), which is a molecule of maternal origin.





TOM: (Translocation outer membrane), the outer membrane translocation complex.

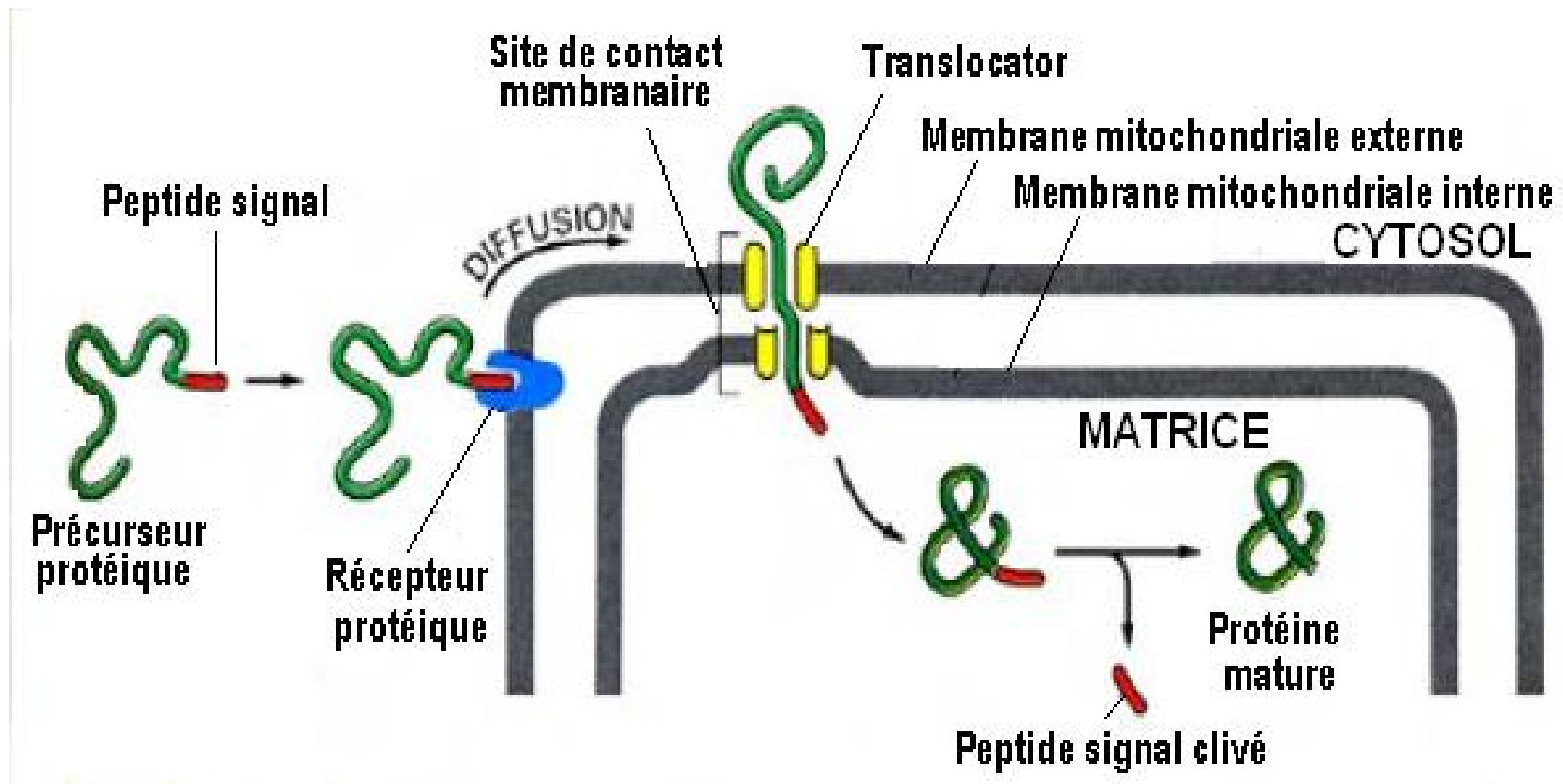
TIM: (Translocation inter membrane), the inner membrane translocation complex.

The receptor complex: plays a role in recognizing the targeting sequences of proteins destined for the mitochondria.

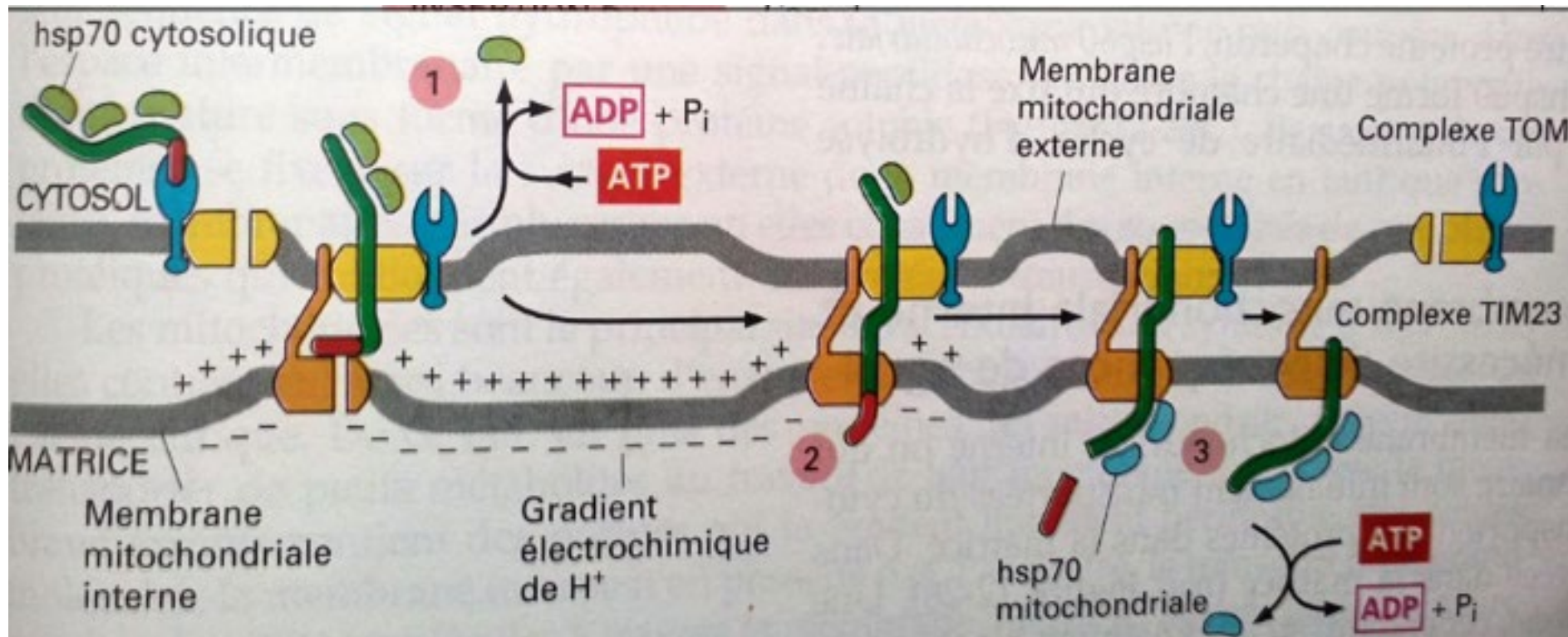
Proteases: participate in protein maturation in the matrix by cleaving the targeting sequences.

Importation des protéines cytosolique dans la mitochondrie

Importation of cytosolic proteins into the mitochondria



1. A cytosolic Hsp70 protein triggers the unwinding of the protein to be imported, allowing it to cross both the TOM and TIM complexes.
2. The targeting sequence is recognized by a protein receptor on the outer membrane.
3. The receptor triggers the translocation of the protein through the TOM protein complex on the outer membrane.
4. The protein translocation continues through a second import complex on the inner membrane, TIM.
5. Proteins that have entered the matrix undergo maturation:
 - A. Cleavage of the targeting sequence by proteases.
 - B. The imported protein assumes its final conformation through the action of Hsp60 and Hsp10 proteins.



Note:

Translocation into mitochondria is driven by both ATP hydrolysis and an electrochemical gradient of H^+ across the inner membrane.

- . Cytosolic hsp70 proteins initiate translocation by hydrolyzing ATP.
- . Mitochondrial (matrix) hsp70 proteins pull the protein during translocation towards the matrix.

HSP means: the heat shock proteins.

Physiological roles:

1/ Calcium storage:

Mitochondria, along with the smooth endoplasmic reticulum, are the main calcium reservoir. They are able to capture calcium, store it in the matrix, and then release it into the cytosol via ion channels in the inner membrane and Na^+/Ca^{++} exchangers.

2/ The respiratory chain: This is the set of protein complexes responsible for oxidative phosphorylation. The transfer of electrons along the complexes of the respiratory chain allows the pumping of protons across the inner membrane. ATP synthesis is coupled to electron transfer via the proton gradient.

3/ Synthesis and Export of Steroid Hormones:

This phenomenon is observed in endocrine cells specialized in the synthesis of steroid hormones.

The mitochondrion cooperates with the smooth endoplasmic reticulum (SER) in the synthesis of these hormones.

The inner mitochondrial membrane carries a family of transmembrane enzymes, the cytochrome P450s, whose active site is bathed in the matrix.

Other enzymes belonging to this family are anchored in the SER membrane, whose active site is bathed in the cytosol.

These enzymes are hydroxylases, which use O₂ and electrons from NADPH (matrix or cytosolic NADPH from pentose phosphate groups) to hydroxylate cholesterol molecules into pregnenolone in the mitochondrial matrix.

Pregnenolone leaves the mitochondrion and reaches the cytosolic side of the SER membrane.

The cytochrome P450 enzymes of the smooth endoplasmic reticulum (SER) synthesize two types of pregnenolone derivatives:

1. Sex hormones (sex steroids).
2. Intermediate metabolites that return to the mitochondrial matrix, where other cytochrome P450 enzymes in the inner membrane then use them for the synthesis of cortisol and aldosterone.

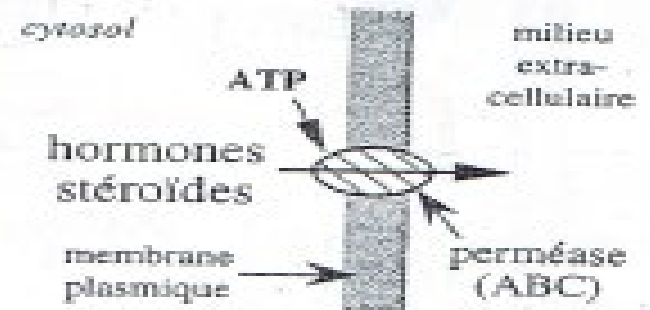
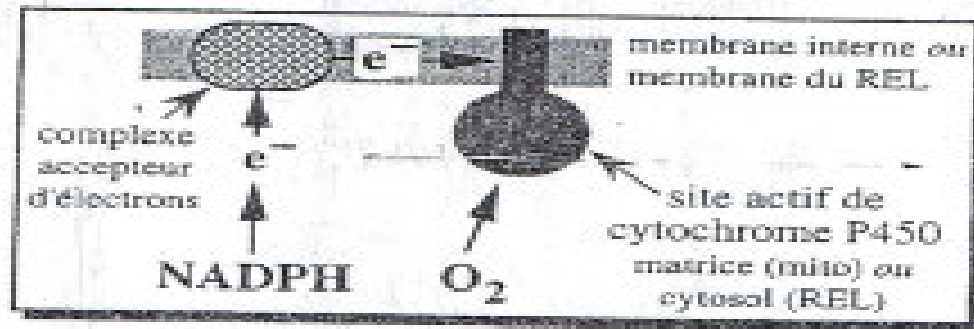
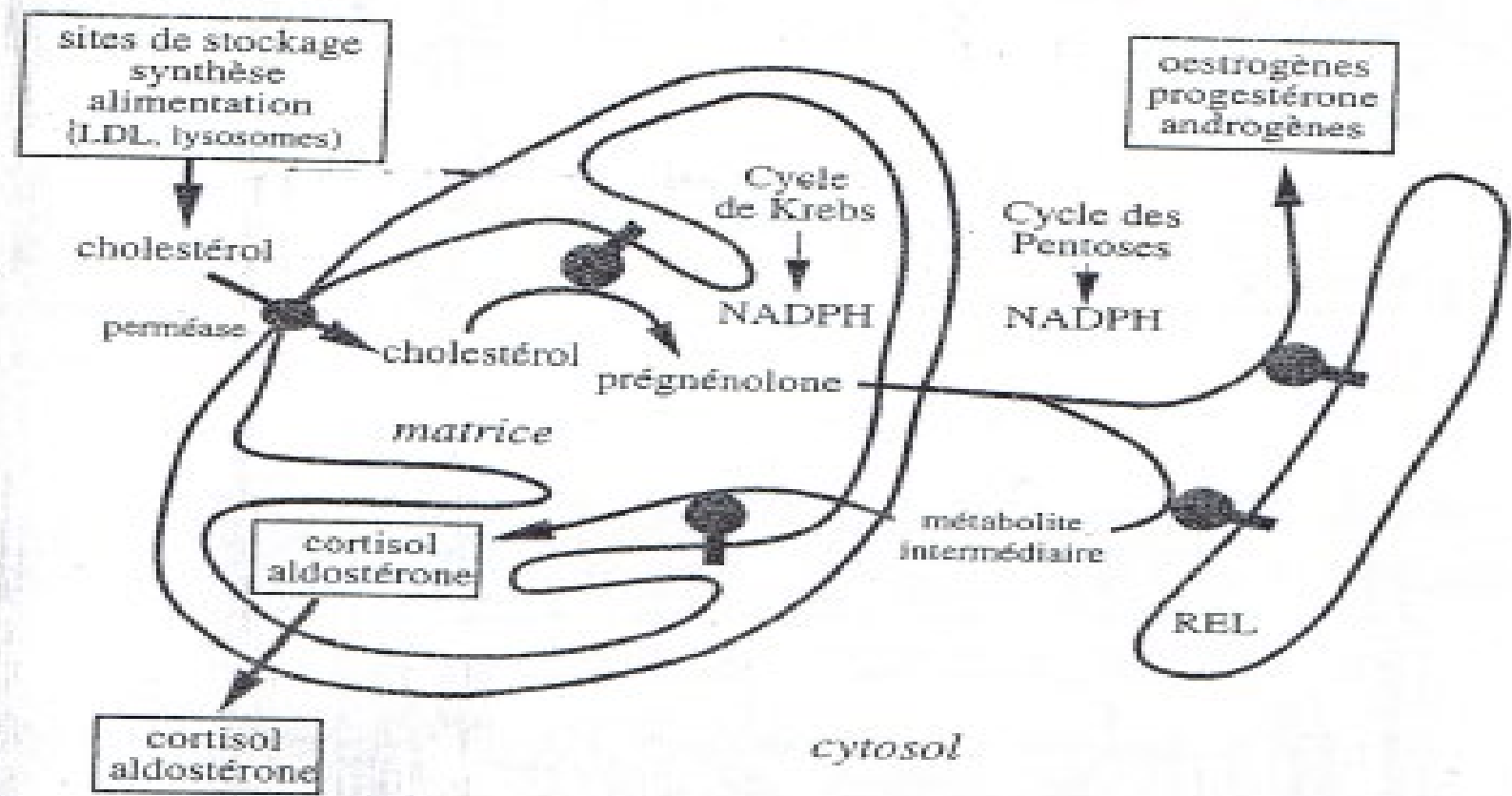


Figure 7 : synthèse et exportation des hormones stéroïdes : un exemple de coopération entre REL, mitochondrie et plusieurs enzymes de la famille des cytochromes P450.

4/ **Protein synthesis under the control of the mitochondrial genome:** Mitochondrial DNA contains 37 genes that code for 13 proteins, 22 transfer RNAs, and 2 ribosomal RNAs. All the encoded proteins are subunits of enzyme complexes of the oxidative phosphorylation system.

